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Frage 1

Vollständig

Erreichte Punkte 1,00 von 1,00

By selecting "I confirm", I hereby declare under oath that I will work on this examination on my own without any help or any third-party assistance.

By selecting "I confirm", I understand that noncompliance results in the invalidation of the assessment, whereby the invalidated examination will be added to the total number of retakes and noncompliance may result in further legal action.

- ☒ a. I confirm
- ☐ b. I do not confirm

Frage 2

Vollständig

Erreichte Punkte 0,00 von 1,00

Which of the following statements about multiple instance learning (MIL) is/are true?

- ☒ a. MIL involves training on sets or bags of instances where each instance is individually labelled.
- ☐ b. MIL requires the set/bag size to be fixed.
- ☐ c. MIL involves training on sets or bags of instances that have a single label.
- ☒ d. MIL models should be invariant to the permutation of instances.

Frage 3

Vollständig

Erreichte Punkte 1,00 von 1,00

Which of the following statements about generative models for small molecules is/are true?

- ☐ a. As the syntax of SMILES strings is complicated and with long dependencies, generators SMILES representations will produce less than 10% valid SMILES strings.
- ☐ b. Generative models are used to explore the known chemical space of synthesized molecules.
- ☐ c. Generative adversarial networks cannot be used to generate small molecules.
- ☒ d. Autoregressive models can be used to generate SMILES representations of small molecules.

Frage 4

Vollständig

Erreichte Punkte 1,00 von 1,00

Which of the following statements about Frechet ChemNet Distance (FCD) and its calculation is/are true?

- ☐ a. FCD measures the distance between individual atoms of molecules.
- ☒ b. FCD is calculated by comparing the means and covariances of the latent representations of molecules.
- ☐ c. FCD is calculated directly on the raw molecular structures without any transformations.
- ☒ d. FCD is a metric used to compare the distribution of generated molecules to that of real molecules.

Frage 5

Vollständig

Erreichte Punkte 0,00 von 1,00

Multi-step retrosynthesis can be formulated as a game that can be played with reinforcement learning. Which of the following statements about the "retrosynthesis game" is/are true?

- ☒ a. The actions in the retrosynthesis "game" involve selecting reactants for the reactions.
- ☐ b. The state in the retrosynthesis "game" refers to the set of available building blocks.
- ☐ c. The environment in the retrosynthesis "game" performs the reaction and returns new molecules.
- ☒ d. The reward in the retrosynthesis "game" is obtained when the product is successfully synthesized from the building blocks.

Frage 6

Vollständig

Erreichte Punkte 1,00 von 1,00

Which of the following statements about AI for chemical reactions is/are true?

- ☒ a. Some methods rely on so-called reaction templates, which are transformation rules for molecules.
- ☒ b. Deep learning methods that predict outcomes of chemical reactions can be trained on databases containing known chemical reactions.
- ☒ c. Transformer architectures have been used to predict outcomes of chemical reactions.
- ☐ d. Predicting chemical reactions with machine learning requires data from quantum mechanical calculations due to bond changes and electron movements.

Frage 7

Vollständig

Erreichte Punkte 1,00 von 1,00

Which of the following statements about the U-Net architecture are correct?

- ☐ a. The U-Net was the first vision model where the encoder is based on self-attention instead of convolutions.
- ☒ b. The U-Net architecture uses a symmetrical encoder and decoder.
- ☐ c. The U-Net has been specifically developed for the object detection task.
- ☒ d. To aid the decoding process, feature maps from the encoder are concatenated with feature maps of the corresponding up-sampling level of the decoder.

Frage 8

Vollständig

Erreichte Punkte 1,00 von 1,00

Assume we are given a table consisting of patient medical records with the following three columns: "smoking status", "cholesterol level", and "diabetes risk score". Entries in the column "smoking status" are given as one of three possible string values (non-smoker, former smoker, current smoker), entries in "cholesterol level" are floating-point measurements indicating the cholesterol level in mg/dL, and entries in "diabetes risk score" are integer values in the range [0-10] indicating the likelihood of a patient developing diabetes. Which statements are correct?

- ☒ a. Missing values in "cholesterol level" can be replaced by the column's median value to handle skewness.
- ☐ b. Column "smoking status" is continuous, as it captures the various smoking habits of patients.
- ☐ c. "cholesterol level" is ordinal since cholesterol measurements can be arranged in increasing order.
- ☐ d. Missing values in "smoker status" can be replaced by the column's mean value to handle skewness.

Frage 9

Vollständig

Erreichte Punkte 1,00 von 1,00

Which of the following statements about ligand-based virtual screening is/are correct?

- ☒ a. Molecular fingerprints are commonly used in ligand-based virtual screening.
- ☐ b. Ligand-based virtual screening is not suitable for predicting the bioactivity of unknown compounds.
- ☐ c. Ligand-based virtual screening requires a 3D structure of the target protein.
- ☐ d. Ligand-based virtual screening typically only considers bioactivity predictions to select compounds.

Frage 10

Vollständig

Erreichte Punkte 1,00 von 1,00

Which of the following statements about medical imaging modalities is/are true?

- ☐ a. Computed Tomography (CT) scans use ultrasounds images to generate images of the body.
- ☒ b. X-rays are used to differentiate between tissues of varying densities, such as bone and soft tissue.
- ☒ c. Magnetic Resonance Imaging (MRI) provides detailed images of internal structures by using strong magnetic fields and radio waves.
- ☐ d. Optical coherence tomography imaging uses multiple X-rays to create images of organs and tissues.

Frage 11

Vollständig

Erreichte Punkte 0,33 von 1,00

Which of the following is the most common task in medical imaging compared to typical vision tasks?

- ☐ a. Image generation
- ☒ b. Classification
- ☒ c. Segmentation
- ☒ d. Object detection

Frage 12

Vollständig

Erreichte Punkte 0,50 von 1,00

Which of the following statements about batch effects in medical data is/are true?

- ☒ a. Ignoring batch effects can lead to models that distinguish between batches rather than the actual biological conditions.
- ☐ b. Batch effects can be eliminated by simply increasing the sample size.
- ☐ c. Batch effects can create artificial differences in data due to variations in experimental conditions.
- ☐ d. Batch effects are usually beneficial as they introduce necessary variability into the data.

Frage 13

Vollständig

Erreichte Punkte 1,00 von 1,00

Which of the following statements about molecular graphs and graph neural networks is/are true?

- ☒ a. A molecular graph can be represented by two matrices: one storing the node features and another one storing the edges between the nodes.
- ☒ b. Node features can include atomic number, hybridization state, and formal charge.
- ☐ c. In a GNN only the node-features are used to make the final predictions.
- ☐ d. The information about the bond type (e.g. single bond, double bond) cannot be included in a GNN.

Frage 14

Vollständig

Erreichte Punkte 1,00 von 1,00

What is the "Clever Hans" effect in the context of AI and medical imaging?

- ☐ a. A bias occurring when enhancing the resolution of low-quality medical images.
- ☒ b. The situation where an AI model appears to perform well but is actually relying on irrelevant features or artifacts in the data.
- ☐ c. The phenomenon where an AI system incorrectly predicts outcomes based on biased training data.
- ☐ d. The phenomenon where data augmentation to improve the robustness of AI models in medical imaging.

Frage 15

Vollständig

Erreichte Punkte 0,67 von 1,00

Which of the following statements about the limitations of GNNs in the context of molecular modeling is/are true?

- ☐ a. GNNs are incapable of predicting any molecular properties accurately.
- ☒ b. GNNs can have difficulty capturing long-range interactions between atoms.
- ☐ c. GNNs may struggle with very large and complex molecular graphs.
- ☒ d. GNNs require a significant amount of computational resources for training.

Frage 16

Vollständig

Erreichte Punkte 1,00 von 1,00

Which of the following statements regarding the characteristics of QSAR/Bioactivity data is/are accurate?

- ☐ a. QSAR datasets usually have data points in the order of thousands or millions.
- ☐ b. QSAR datasets typically have a balanced distribution of active and inactive compounds.
- ☒ c. QSAR datasets often contain series of structurally related compounds.
- ☒ d. QSAR data often includes noise due to biological variability and experimental inconsistencies.

Frage 17

Vollständig

Erreichte Punkte 0,50 von 1,00

Data from the medical and healthcare sectors about patients is often affected by domain shifts. Which of the following statements about domain shifts in medical and healthcare data are true?

The usual notation of the lecture is used, where the random variable (y) represents the disease status (e.g. "healthy" and "diseased") and ($\mathbf{x}$) represents the patient features, e.g., age, blood parameters, or heart rate.

- ☐ a. If a machine learning model has a high predictive quality on a randomly drawn validation set, this means it will automatically adjust to domain shifts and maintain predictive quality over time.
- ☐ b. The distribution of the patient features, that is the input data ($\mathbf{x}$) with distribution ($p(\mathbf{x})$) for machine learning models, is never affected by domain shifts.
- ☒ c. During disease waves of infectious diseases (such as COVID-19), the prevalence of a disease and thus the probability of observing this phenotype ($p(\text{👉})$) changes.
- ☐ d. Assume that the diagnostic process to determine the disease status (y) is changed, for example, an antigen test is replaced by a PCR test. This represents a concept shift ($p(y|\mathbf{x})$).

Frage 18

Vollständig

Erreichte Punkte 1,00 von 1,00

Which of the following statements about SMILES is/are correct?

- ☒ a. SMILES stands for Simplified Molecular Input Line Entry System.
- ☒ b. SMILES strings represent molecules in a linear sequence of characters.
- ☐ c. SMILES cannot be converted into a molecular graph.
- ☐ d. SMILES is always unique for each molecule, ensuring no two different molecules share the same SMILES.

Frage 19

Vollständig

Erreichte Punkte 0,67 von 1,00

Which of the following statements about ADME (Absorption, Distribution, Metabolism, and Excretion) properties is/are true?

- ☒ a. Lipinski's rule of five can be used a proxy for ADME properties.
- ☐ b. Good ADME properties guarantee that a compound will be an effective drug.
- ☐ c. ADME properties such as clearance are often difficult to model.
- ☒ d. ADME properties are critical in determining the pharmacokinetics of a drug.

Frage 20

Vollständig

Erreichte Punkte 1,00 von 1,00

Which of the following applications is/are relevant for QSAR modeling?

- ☐ a. Creating 3D protein structure models.
- ☒ b. Identifying potential drug candidates.
- ☒ c. Predicting the toxicity of new chemical compounds.
- ☐ d. Mapping the genome sequences of organisms.

Frage 21

Vollständig

Erreichte Punkte 0,50 von 1,00

Which of the following statements about multimodal contrastive learning models is/are true?

- ☐ a. Multi-modal contrastive learning encodes multiple modalities (e.g. molecules and proteins) into a shared latent space.
- ☒ b. Multi-modal contrastive learning is a self-supervised learning technique.
- ☐ c. Multi-modal contrastive learning concentrates (latent) representations from multiple modalities (e.g. molecules and proteins) to make predictions.
- ☐ d. Multi-modal contrastive learning is an unsupervised learning technique.

Frage 22

Vollständig

Erreichte Punkte 0,50 von 1,00

In virtual screening, the quality of the top-ranked predictions is crucial for efficient drug discovery. Which of the following methods are specifically adapted to prioritize high-quality early hits in the screening process?

- ☐ a. Precision-Recall Curve
- ☒ b. BEDROC (Boltzmann-Enhanced Discrimination of ROC)
- ☐ c. Average Precision (AP)
- ☐ d. Enrichment Factor (EF)

Frage 23

Vollständig

Erreichte Punkte 1,00 von 1,00

Which of the following statements about synthesis planning is/are true?

- ☒ a. Synthesis planning involves designing a sequence of chemical reactions to produce a target compound.
- ☒ b. Retrosynthetic analysis is a key technique used in synthesis planning to break down complex molecules into simpler starting materials.
- ☐ c. Synthesis planning has been made trivial by modern deep learning methods.
- ☐ d. Synthesis planning involves only selecting the reactants.

Frage 24

Vollständig

Erreichte Punkte 1,00 von 1,00

Which of the following statements about synergy/antagonism is/are true?

- ☒ a. Drug synergy occurs when the combined effect of two or more drugs is greater than the sum of their individual effects.
- ☒ b. The Loewe additivity model is commonly used to assess drug synergy.
- ☐ c. Antagonistic effects occur when the combined effect of two or more drugs is greater than the sum of their individual effects.
- ☐ d. Drug synergy/antagonism evaluation is only applicable to chemically similar drugs.

Frage 25

Vollständig

Erreichte Punkte 1,00 von 1,00

Which of the following statements about goal-directed molecule generation is/are true?

- ☐ a. Goal-directed generation focuses only on replicating existing molecular structures with desired properties.
- ☐ b. Goal-directed generation aims to create molecules with similar properties to the training set.
- ☒ c. Reinforcement learning can be used to optimize goal-directed molecule generation.
- ☒ d. Goal-directed generation aims to create molecules with specific desired properties.

Frage 26

Vollständig

Erreichte Punkte 0,50 von 1,00

Which of the following statements about data splits in QSAR model evaluation is/are true?

- ☐ a. Temporal splits help evaluate the model's performance in future, unseen data by simulating real-world data acquisition over time.
- ☐ b. Generally, random split is the best method for evaluating QSAR models as it eliminates biases.
- ☐ c. Cross-validation is rarely used in the evaluation of QSAR due to computational cost.
- ☒ d. Cluster splits group similar compounds together, ensuring that structurally similar compounds are not both in the training and test sets.

Frage 27

Vollständig

Erreichte Punkte 1,00 von 1,00

Which of the following statements about target identification/deconvolution is/are correct?

- ☒ a. For a given molecule, it consists of identifying to which protein it will most likely bind if multiple are present simultaneously.
- ☒ b. Multi-class classification can be used for it.
- ☐ c. Multi-task classification can be used for it.
- ☐ d. For a given molecule, it consists of identifying all the proteins it will bind if multiple are present simultaneously.

Frage 28

Vollständig

Erreichte Punkte 0,50 von 1,00

Which of the following statements about AlphaFold 2 are true?

- ☐ a. AlphaFold 2 incorporates quantum mechanical calculations to improve the accuracy of its predictions.
- ☒ b. AlphaFold 2 incorporates multiple sequence alignments and homologous protein structures to improve the accuracy of its predictions.
- ☒ c. AlphaFold 2 is a deep learning model designed to predict the three-dimensional structure of proteins from their amino acid sequences.
- ☒ d. AlphaFold 2 uses a convolutional neural network architecture.

Frage 29

Vollständig

Erreichte Punkte 1,00 von 1,00

Which of the following statements about bioassays is/are true?

- ☒ a. They can be used to measure the biological response or function elicited by a substrate.
- ☒ b. They are typically test the substrate at various concentrations.
- ☒ c. They are wet-lab procedures to acquire knowledge of a compound in a biological context.
- ☐ d. They are typically tested *in vitro*, i.e. in a living organism.

Frage 30

Vollständig

Erreichte Punkte 0,00 von 1,00

Which of the following statements about explainable AI and interpretability in medical imaging is/are true?

- ☒ a. Gradient integration utilizes the propagation of a relevance value back to input layers by layer to estimate the relevance of each input feature.
- ☒ b. Explainable AI techniques, such as Grad-CAM, help visualize which parts of a medical image are most important for a model's prediction.
- ☒ c. Occlusion analysis involves removing patches of the training data images to improve the model's interpretability.
- ☒ d. Counterfactual explanations in medical imaging involve generating alternative images to show what changes would have been needed for a different diagnosis.

Frage 31

Vollständig

Erreichte Punkte 0,50 von 1,00

Which of the following statements about molecular fingerprints is/are correct?

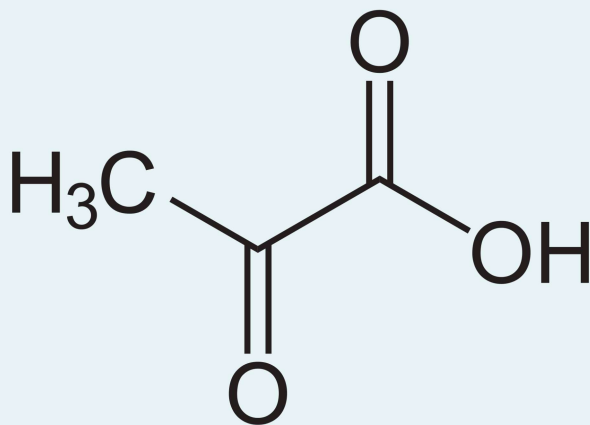
- ☐ a. A unique molecular graph can be constructed from them.
- ☐ b. They are binary or numerical representations of molecular structures.
- ☒ c. They are used for similarity searching and machine learning tasks.
- ☐ d. They provide a 3D representation of molecules.

Frage 32

Vollständig

Erreichte Punkte 0,50 von 1,00

Given the following molecule:



Which of the following SMILES strings encodes this molecule?

- ☐ a. O=C(C(=O)O)C
- ☒ b. CC(=O)C(=O)O
- ☐ c. CCCCOO
- ☐ d. OCCOO

Frage 33

Vollständig

Erreichte Punkte 1,00 von 1,00

Which of the following statements about Deep Learning techniques used for analyzing biological sequences is/are true?

- ☒ a. Feature extraction from biological sequences can be used to identify motifs, which are recurring patterns or elements within the sequences.
- ☒ b. One-hot encoding is commonly used to prepare biological sequences as input for deep learning models.
- ☒ c. Deep Learning techniques can be applied to tasks such as classification, segmentation, and generative modeling of biological sequences.
- ☒ d. 1D-Convolutions, Recurrent Neural Networks (RNNs), Long Short-Term Memory Networks (LSTMs), and Transformers are all used to analyze biological sequences.

Frage 34

Vollständig

Erreichte Punkte 1,00 von 1,00

Which of the following statements about the use of Random Forest (RF) and Graph Neural Networks (GNNs) is/are true?

- ☐ a. GNNs are inherently better for QSAR modeling because they naturally capture the graph-like structure of molecules, leading to more accurate predictions in all cases.
- ☒ b. RF models are less prone to overfitting compared to GNNs, making them a more reliable choice for datasets with a large number of features but limited samples.
- ☒ c. RF models are typically more interpretable than GNNs, making it easier to understand the contribution of individual molecular features to the model's predictions.
- ☐ d. GNNs are less sensitive to the quality and variability of the input data compared to RFs, making them more consistent in performance across different datasets.

Frage 35

Vollständig

Erreichte Punkte 1,00 von 1,00

Which of the following statements about the composition of biological sequences is/are true?

- ☒ a. Protein sequences are composed of chains of amino acids linked by peptide bonds.
- ☒ b. DNA sequences are composed of nucleic acids.
- ☐ c. Biological sequences do not follow any specific patterns or rules in their composition.
- ☐ d. RNA sequences are composed of lipids.